

INSTITUTE OF SPACE TECHNOLOGY (IST), ISLAMABAD

Department of Space Science · BS Space Science Final Year Project

Multi-Agent Deep Reinforcement Learning for UAV Swarm Navigation in a Geospatial Digital Twin

Project Title:	Multi-Agent Deep Reinforcement Learning for UAV Swarm Navigation in a Geospatial Digital Twin
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Target Simulators:	Blender 4.x / Unreal Engine 5.4 (Chaos Physics)
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Final Year Project Proposal — BS Space Science

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1. Abstract

Autonomous Unmanned Aerial Vehicle (UAV) swarms operating in dense, GPS-degraded, or post-disaster urban environments face severe navigational hazards, including narrow urban canyons, unmapped tree canopies, and suspended power transmission lines. Training cooperative Multi-Agent Deep Reinforcement Learning (MADRL) policies directly in the physical world is prohibitively risky and cost-inefficient. Conversely, conventional synthetic robotics simulators rely on idealized, procedural environments that suffer from an insurmountable "sim-to-real" domain gap due to a total lack of grounding in real-world geographic morphology and physical clutter.

This BS Space Science Final Year Project presents **TerraTwin Swarm**, an end-to-end framework integrating remote sensing, automated 3D geometry reconstruction, and reinforcement learning. The system reconstructs a high-fidelity geospatial digital twin from sub-meter commercial satellite imagery (SpaceNet 2, Las Vegas) over a bounded urban corridor. The extraction methodology leverages a three-branch pipeline:

1. **Branch A:** Universal Transformer instance segmentation (**Mask2Former**) refined with the **Segment Anything Model 2 (SAM 2)** for sub-pixel boundary snapping, regularized via **Douglas-Peucker orthogonalization** into cadastre-grade polygons;
2. **Branch B:** Ground filtering applied to stereo Digital Surface Models (DSM) to compute Normalized Digital Surface Models (**nDSM**), cross-validated against solar shadow trigonometry ($h = L_{\text{shadow}} \cdot \tan \theta_s$) using ephemeris metadata;
3. **Branch C:** Environmental micro-hazard isolation separating tree canopies via Soil-Adjusted Vegetation Index (**SAVI**) and reconstructing overhead power transmission lines via **catenary curve physics** ($y = a \cdot \cosh(x/a) - a$).

Processed vector assets are optimized through automated Blender geometry pipelines and compiled into **Unreal Engine 5 (UE5)**, establishing an industrial-grade simulation environment equipped with **Chaos Physics colliders**. Within this geospatial twin, a cooperative swarm of quadrotors governed by 6-DOF kinematics and ray-cast LiDAR sensors is trained using **Multi-Agent Proximal Policy Optimization (MAPPO)**. The agents learn decentralized cooperative trajectory planning, collision avoidance, and acoustic beacon localization under severe aerodynamic and geometric constraints.

2. Introduction & Problem Statement

2.1 Context & Motivation

Urban aerial autonomy is transitioning from single, high-altitude UAV missions to distributed, low-altitude multi-UAV swarm operations. Critical applications—including post-seismic search and rescue (SAR), emergency medical payload delivery, and urban surveillance—require swarms to navigate through complex urban corridors below the building roofline. In these environments, standard Global Navigation Satellite System (GNSS) signals suffer from severe multipath interference, urban canyon blackouts, and complete loss of carrier lock.

To achieve autonomy without continuous GNSS availability, swarms must rely on onboard sensors (LiDAR, optical odometry) and learned spatial behaviors. MADRL offers a powerful framework for discovering decentralized cooperative policies: agents communicate locally, avoid mutual collisions, and coordinate search sweeps without requiring a central coordinator.

However, training MADRL algorithms requires millions of environmental interaction steps. Doing so on physical hardware results in frequent catastrophic crashes and hardware destruction. Simulators are therefore essential. Yet existing simulators fall into two unsatisfactory extremes:

- **Robotics simulators (Gazebo, Webots):** Possess physics engines but rely on flat planes and simple geometric primitives (boxes, cylinders), lacking real-world geographic complexity.
- **Procedural game worlds:** Provide visual fidelity but possess no semantic or geographic correspondence to any real place on Earth.

Geospatial Digital Twins—digital replicas of real-world physical environments generated from Earth observation data—bridge this divide.

2.2 Problem Statement

Currently, there exists **no automated, reproducible engineering pipeline** that:

1. Ingests raw sub-meter optical satellite imagery and satellite metadata;
2. Extracts high-precision building footprints, structural heights, and low-altitude micro-hazards (catenary powerlines, tree canopies);
3. Compiles these features into an industrial-grade physics engine (**Unreal Engine 5**) with accurate collision meshes; and
4. Implements and evaluates a closed-loop **Multi-Agent Deep Reinforcement Learning (MADRL)** swarm navigation policy directly inside the reconstructed geospatial digital twin.

2.3 Research Questions

1. **Extraction Precision:** How effectively can transformer-based foundation models (Mask2Former + SAM 2) resolve sharp building corners and structural boundaries in dense satellite imagery compared to traditional convolutional U-Nets?

-
2. **Topographic & Shadow Validation:** Can monocular satellite shadow geometry ($h = L_{\text{shadow}} \cdot \tan \theta_s$) successfully bound elevation estimation errors in the absence of dense airborne LiDAR ground truth?
 3. **Micro-Hazard Modeling:** What is the minimum physical representation of catenary transmission lines and vegetative canopies necessary to train collision-free low-altitude UAV flight paths?
 4. **MADRL Swarm Convergence:** How does training under 6-DOF kinematic constraints in a topologically faithful digital twin impact MAPPO convergence speed, inter-agent collision rates, and target localization efficiency relative to simplified Euclidean obstacle environments?
-

3. Aims & Measurable Objectives

3.1 Aim

To design, develop, and evaluate an automated pipeline that reconstructs a high-fidelity 3D geospatial digital twin from optical satellite imagery and trains an autonomous, cooperative UAV swarm using Multi-Agent Deep Reinforcement Learning (MADRL) for safe low-altitude urban navigation.

3.2 Measurable Objectives (Mapped to Work Breakdown Structure)

ID	Project Stage	Objective Description	Target Metric / Acceptance Criterion
O1	Stage 1 (Feature Extraction)	Extract building footprint instances from SpaceNet 2 using Mask2Former + SAM 2 boundary refinement.	Mask AP ≥ 0.58 , Boundary IoU ≥ 0.78 on held-out test split.
O2	Stage 2 (Topography & Hazards)	Compute building heights via stereo nDSM and validate via solar shadow trigonometry; map vegetation (SAVI) and catenary powerlines.	Building height MAE ≤ 1.8 m; catenary sag clearance accuracy $\geq 90\%$.
O3	Stage 3 (Mesh Compilation)	Vector-to-mesh extrusion in Blender and compilation into Unreal Engine 5 with Chaos Physics colliders.	Watertight manifold mesh ($\geq 99.5\%$ valid); zero non-physical collider penetrations; steady 60 FPS in UE5.
O4	Stage 4 (Swarm Kinematics)	Model 6-DOF quadrotor rigid-body dynamics, ray-cast LiDAR emulation, and acoustic rescue signal attenuation.	Physics stability with multi-rotor drag and rotor inertia; real-time LiDAR point-cloud generation (≥ 10 Hz).
O5	Stage 4 (MADRL Policy)	Train a cooperative UAV swarm using MAPPO for acoustic target search while avoiding terrain, buildings, and catenary hazards.	Target reach rate $\geq 92\%$; mutual collision rate $< 3\%$; zero powerline entanglement incidents.
O6	Benchmarking & Analysis	Conduct ablation studies against baseline heuristic planners (A / RRT) and traditional semantic U-Nets.	Comprehensive quantitative analysis documenting convergence, sim-to-real fidelity, and failure modes.

4. Literature Review & Theoretical Background

4.1 Satellite Semantic Feature Extraction & Foundation Models

Traditional remote sensing feature extraction relied on pixel-level Convolutional Neural Networks (CNNs) such as the U-Net (Ronneberger et al., 2015). While effective for rough land-cover classification, convolutional decoders struggle with spatial sharpness: pooling layers discard high-frequency boundary information, producing rounded corners, merged adjacent structures, and jagged raster staircasing.

Recent breakthroughs in Vision Transformers (ViT) have revolutionized geospatial segmentation. **Mask2Former** (Cheng et al., 2022) introduces masked-attention mechanisms that extract localized feature queries, treating semantic, instance, and panoptic segmentation under a unified framework. Furthermore, the advent of visual foundation models such as **Segment Anything Model 2 (SAM 2)** (Ravi et al., 2024) provides promptable, high-frequency boundary refinement capable of sub-pixel zero-shot boundary tracing.

Combining Mask2Former instance queries as bounding prompts for SAM 2 produces cadastre-grade building boundaries without manual intervention.

4.2 Topographic Extraction, nDSM & Shadow Trigonometry

Height estimation from single optical overhead imagery is fundamentally ill-posed due to projective scale ambiguity. In satellite photogrammetry, building height extraction historically relies on Digital Surface Models (DSM) generated via dense stereo image matching (SGM) or LiDAR point clouds:

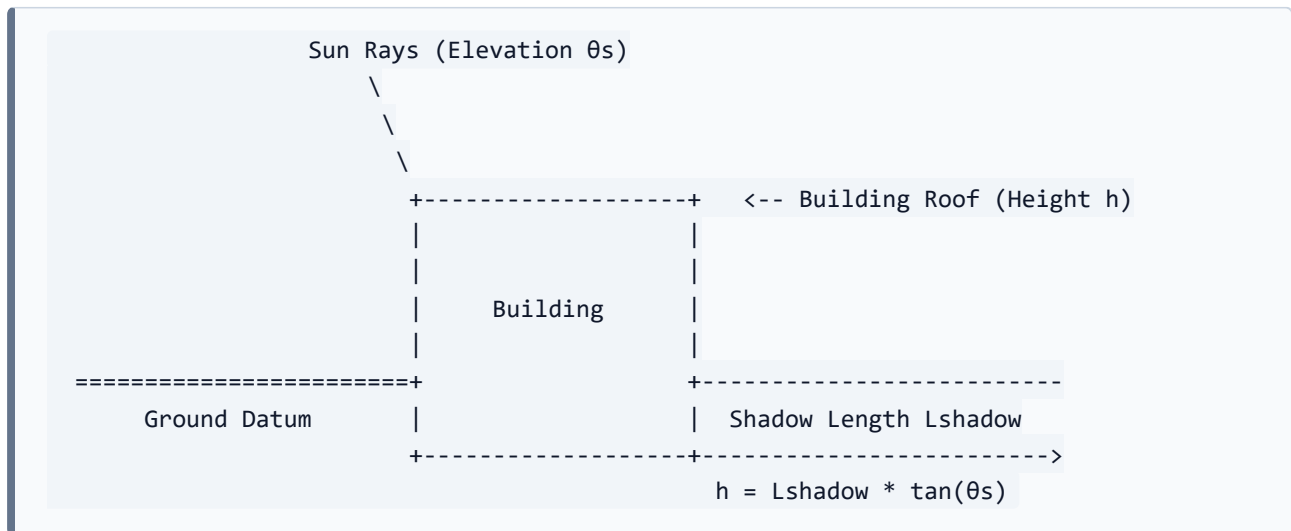
$$\text{nDSM} = \text{DSM} - \text{DTM}$$

where DTM is the Digital Terrain Model representing the bare ground datum.

Where stereo coverage is constrained, **shadow-geometry trigonometry** provides independent, physics-grounded validation. When sun azimuth (ϕ_s) and sun elevation angle (θ_s) are recorded in the satellite metadata XML (DigitalGlobe/Maxar ephemeris), the geometric height (h) of a planar building casting a shadow of length L_{shadow} onto horizontal terrain is governed by:

$$h = L_{\text{shadow}} \cdot \tan(\theta_s)$$

Accounting for the satellite sensor view zenith angle (θ_v) and relative azimuth ($\phi_s - \phi_v$), monocular shadow mensuration provides rigorous mathematical cross-validation against estimated nDSM values.



4.3 Micro-Hazard Identification & Physical Modeling

Low-altitude UAV flight safety is heavily endangered by micro-hazards invisible on coarse land-use maps:

1. **Tree Canopies:** Soft-body collisions damage rotor blades and trigger propeller stall. Canopies are segmented using the Soil-Adjusted Vegetation Index (SAVI) to suppress desert soil background albedo:

$$\text{SAVI} = \frac{(1 + L)(\rho_{\text{NIR}} - \rho_{\text{Red}})}{\rho_{\text{NIR}} + \rho_{\text{Red}} + L}, \quad L = 0.5$$

1. **Transmission Lines & Catenary Wires:** High-voltage overhead cables represent the single greatest hazard to low-altitude aerial robotics. Transmission towers detected along road corridors are connected via physical **catenary curve equations**. A cable suspended between two towers at horizontal distance D with sag constant $a = \frac{T_0}{\mu g}$ (where T_0 is horizontal cable tension and μ is linear mass density) follows:

$$y(x) = a \cdot \left[\cosh\left(\frac{x}{a}\right) - 1 \right]$$

4.4 Industrial-Grade Geospatial Digital Twins (UE5 vs Traditional Simulators)

Traditional robotics platforms (Gazebo, Webots, Bullet Physics) are designed for ground rovers and simple robotic arms. They struggle to render complex geospatial meshes comprising hundreds of thousands of polygons and high-frequency collision hulls.

Unreal Engine 5 (UE5) provides a state-of-the-art simulation foundation:

- **Chaos Physics:** High-performance, deterministic rigid-body and soft-body solver capable of handling hundreds of concurrent convex collision hulls.
- **Nanite Virtualized Geometry:** Allows rendering billions of polygons without manual level-of-detail (LOD) degradation.
- **Large World Coordinates (LWC):** 64-bit double-precision floating-point coordinate space enabling metric georeferenced bounding regions without precision jitter.

4.5 Multi-Agent Deep Reinforcement Learning (MADRL) for UAV Swarms

Decentralized multi-UAV navigation in continuous state-action spaces is formulated as a **Decentralized Partially Observable Markov Decision Process (Dec-POMDP)**, defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \Omega, \mathcal{O}, \mathcal{N}, \gamma \rangle$.

Standard single-agent reinforcement learning (e.g., standard PPO, DDPG) suffers from environmental non-stationarity when multiple agents learn simultaneously. **Multi-Agent Proximal Policy Optimization (MAPPO)** (Yu et al., 2022) implements the **Centralized Training with Decentralized Execution (CTDE)** paradigm:

- **Centralized Critic:** Has access to the global state s_t , evaluating team-wide value estimates during offline training.
- **Decentralized Actor:** Evaluates only the local agent observation o_i^t (onboard LiDAR rays, local telemetry, target acoustic signal gradient) during online execution.

5. System Scope & Bounded Region Definition

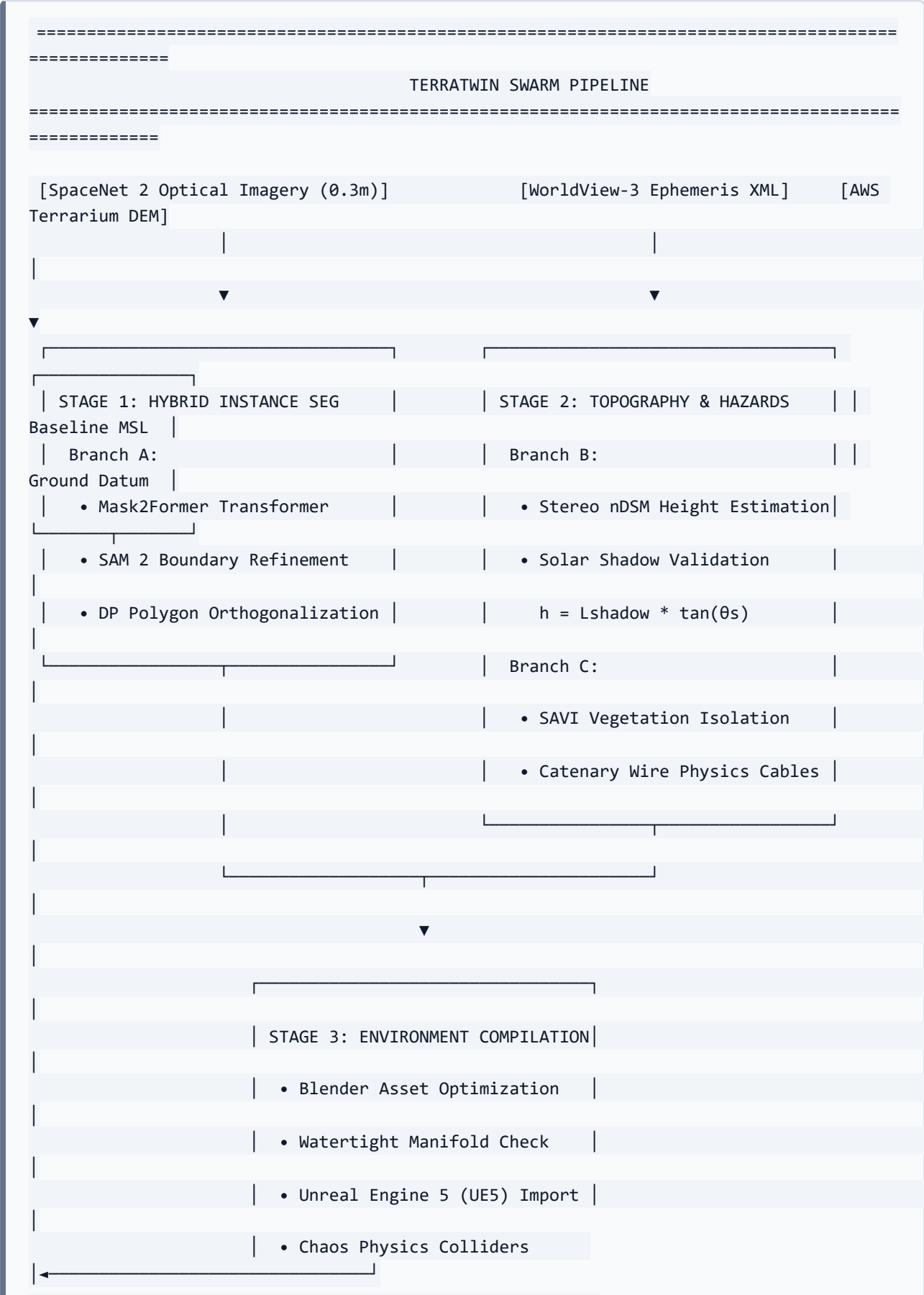
To achieve high scientific rigor while respecting local GPU hardware constraints (GTX 1660 Super / local CUDA environment), the project explicitly bounds its operational geospatial region.

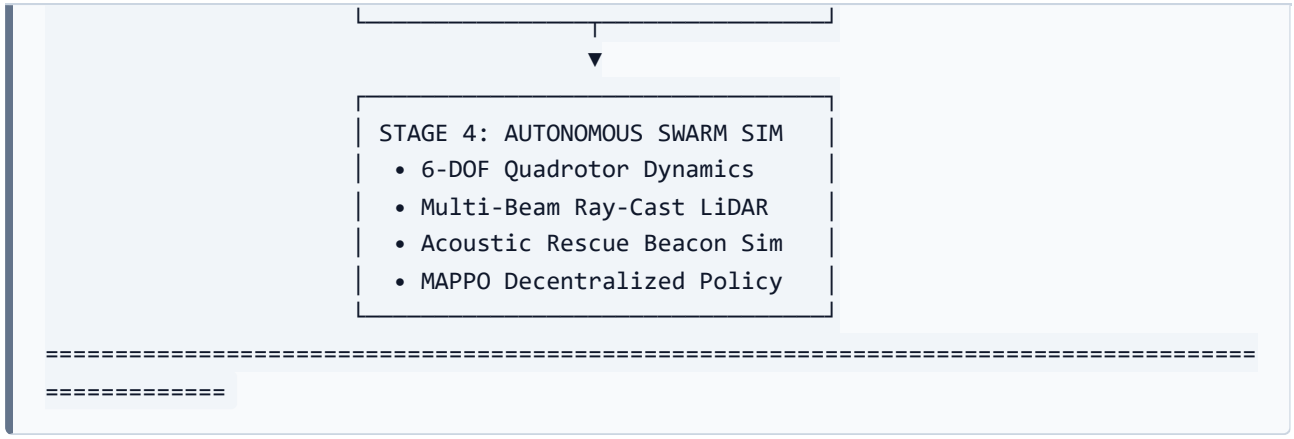
5.1 Area of Interest (AOI): SpaceNet 2 Las Vegas Bounded Corridor

The digital twin is constructed from **SpaceNet 2 (AOI 2 — Las Vegas)** commercial satellite imagery:

- **Sensor:** WorldView-3 optical satellite (DigitalGlobe/Maxar).
 - **Spectral Resolution:** 3-channel RGB-PanSharpened (450–690 nm).
 - **Ground Sampling Distance (GSD):** 0.30 m per pixel nadir resolution.
 - **Bounded Geographic Footprint:** A continuous urban corridor encompassing **Sector 7 (Las Vegas)** covering approximately 4.0 km²:
 - Latitude: 36.1265° N to 36.2599° N
 - Longitude: –115.2777° W to –115.1583° W
 - Structural Population: 1,016 validated building structures + 20 major road segments.
 - **Rationale for Bounded Subsetting:** Restricting the digital twin to a bounded, high-density residential and commercial corridor allows generating full 3D collision geometry at millimeter precision, running high-frequency ray-cast LiDAR simulations in real time, and maintaining 60 FPS in Unreal Engine 5.
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6. Methodology & Pipeline Architecture





6.1 Stage 1 & 2: High-Precision Geospatial & Micro-Hazard Extraction

The extraction architecture operates along three parallel branches:

6.1.1 Branch A: Universal Instance Segmentation (*Mask2Former + SAM 2 + DP Orthogonalization*)

1. **Instance Query Extraction (Mask2Former):** The WorldView-3 0.3m RGB tiles are ingested by a Swin-Transformer-backed **Mask2Former** instance segmenter fine-tuned on urban footprint topologies. Unlike semantic segmentation models that predict a single heat-map, Mask2Former extracts discrete object queries:

$$\mathcal{L}_{\text{Mask2Former}} = \lambda_{\text{cls}} \mathcal{L}_{\text{CE}} + \lambda_{\text{bce}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dice}} \mathcal{L}_{\text{Dice}}$$

1. **Zero-Shot Boundary Refinement (SAM 2):** Bounding boxes from predicted instances initialize prompt queries into **Segment Anything Model 2 (SAM 2)**. SAM 2 processes high-frequency edge gradients to eliminate raster staircasing and snap boundary edges along physical roof gutters.
2. **Douglas-Peucker Orthogonalization:** The refined contours are vectorized via Shapely/GDAL and regularized using a modified Douglas-Peucker algorithm enforcing $90^\circ \pm \epsilon$ orthogonality on urban corners:

$$\epsilon_{\text{ortho}} = \min_{\theta \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}} |\theta_{\text{segment}} - \theta|$$

This eliminates non-physical jagged vertices while preserving true footprint area.

6.1.2 Branch B: Topography, nDSM & Shadow Trigonometric Validation

1. **Normalized Digital Surface Model (nDSM):** Digital Surface Models derived from stereo pairs are filtered using morphological opening to construct the bare-earth DTM. Building elevation is isolated via:

$$z_{\text{building}} = \text{DSM}(x, y) - \text{DTM}(x, y)$$

1. **Shadow Trigonometric Cross-Validation:** For every segmented structure, the sun elevation angle θ_s and azimuth ϕ_s are parsed from the tile's metadata XML. The shadow profile is extracted along the anti-solar vector $\vec{v}_{\text{shadow}} = [-\sin \phi_s, -\cos \phi_s]$. The calculated height $h_{\text{shadow}} = L_{\text{shadow}} \cdot \tan \theta_s$ is used to validate and constrain the nDSM model:

$$h_{\text{final}} = \alpha \cdot h_{\text{nDSM}} + (1 - \alpha) \cdot h_{\text{shadow}}, \quad \alpha \in [0.7, 0.9]$$

6.1.3 Branch C: Micro-Hazard Mapping (NDVI/SAVI Canopies & Catenary Powerlines)

1. **Vegetation Extraction:** In arid/semi-arid urban zones, background soil brightness skews traditional NDVI. The Soil-Adjusted Vegetation Index (SAVI) with soil correction parameter $L = 0.5$ is computed across available multispectral bands. Clusters with $\text{SAVI} > 0.28$ are isolated, extruded as volumetric vegetative obstacle cylinders, and flagged with soft-body aerodynamic turbulence penalties.
2. **Transmission Lines & Catenary Cables:** Power utility corridors along roadways are identified. Towers serve as structural anchors. Cable sag profiles are computed via catenary curves:

$$z_{\text{cable}}(x) = z_0 + a \left[\cosh \left(\frac{x - x_{\text{mid}}}{a} \right) - 1 \right]$$

These cables are compiled as continuous lethal collision cylinders in the flight simulator, directly addressing an urban obstacle that is completely absent from all standard datasets.

6.2 Stage 3: Environment Compilation & Unreal Engine 5 Physics Sim

6.2.1 Vector-to-Mesh Bridge & Blender Asset Optimization

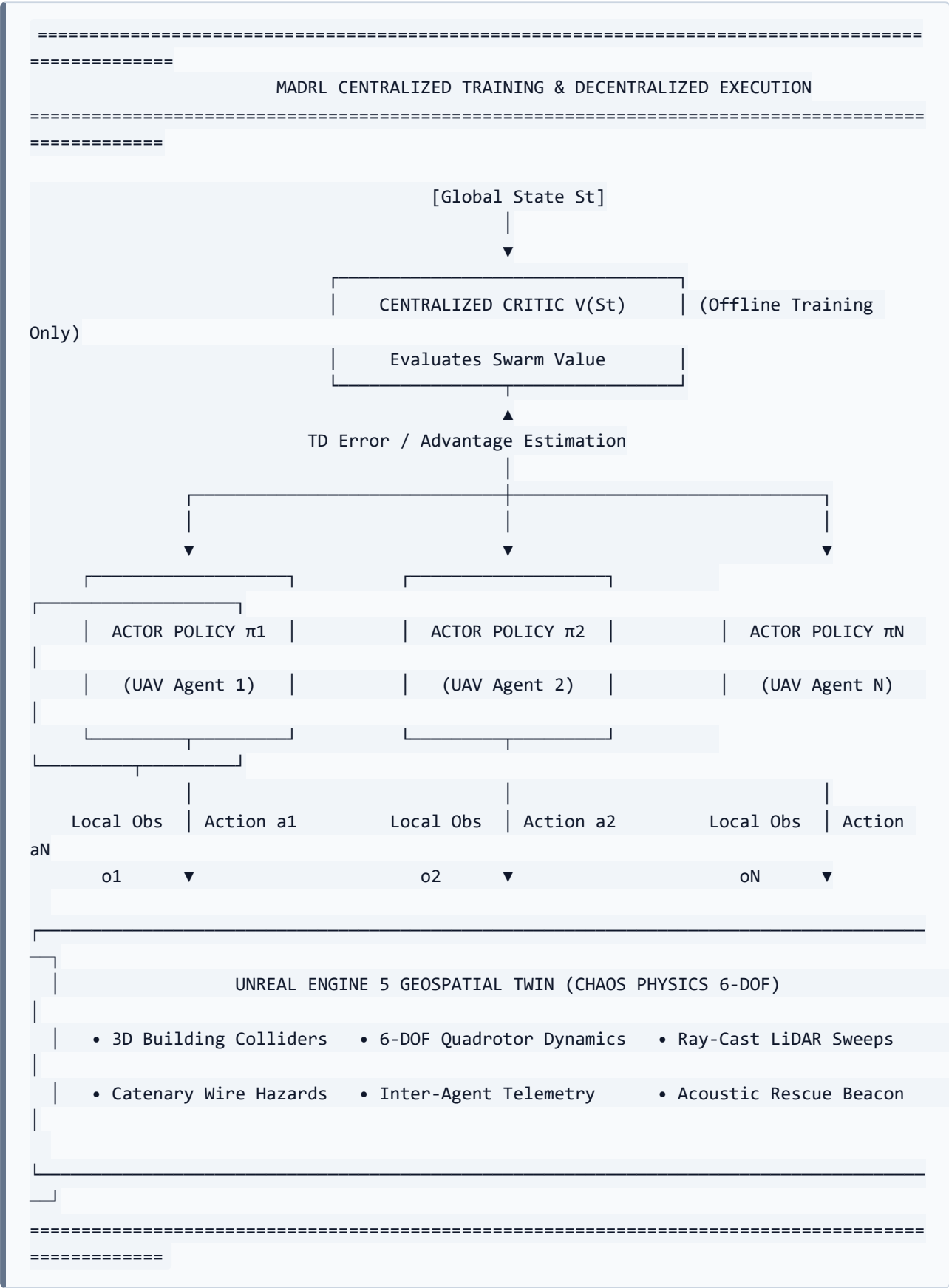
Raw GIS polygons cannot be imported directly into a real-time game engine without preprocessing:

1. **Automated Blender Pipeline (bpy headless):** A Python automation script in Blender ingests the 2.5D GeoJSON footprints, structural heights, and catenary lines.
2. **Triangulation & Watertight Validation:** Roof faces are triangulated using constrained Delaunay triangulation (earcut); vertical walls are extruded down to the DEM ground datum. Every mesh is verified manifold (zero non-manifold edges, zero internal intersecting faces).
3. **Decimation & LODs:** Planar walls are simplified to minimize vertex count while preserving collision boundaries.

6.2.2 Unreal Engine 5 Ingestion & Chaos Physics Colliders

1. **Nanite & Landscape:** The AWS Terrarium DEM ground datum is imported as a 16-bit displacement heightmap. The ground surface is textured with high-resolution satellite imagery.
 2. **Chaos Physics Convex Decomposition:** Extruded buildings are converted into static rigid bodies equipped with **Chaos Physics Convex Hull Colliders**.
 3. **Lethal Wire Volumes:** Catenary cables are instantiated as thin, non-deformable capsule collision volumes capable of triggering immediate UAV rotor destruction upon intersection.
-

6.3 Stage 4: Autonomous UAV Swarm Engine (MADRL Framework)



6.3.1 6-DOF Quadrotor Dynamics & Flight Envelope

Each UAV in the swarm is modeled as a 6-DOF rigid body under aerodynamic forces:

$$\dot{\mathbf{p}} = \mathbf{v}$$

$$m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{R}\mathbf{F}_{\text{thrust}} - \mathbf{D}\mathbf{v}$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega}]_{\times}$$

$$\mathbf{I}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})$$

where:

- $\mathbf{p} = [x, y, z]^T$ is the world position;
- $\mathbf{v}$ is the linear velocity vector;
- $\mathbf{R} \in SO(3)$ is the rotation matrix from body to world frame;
- m is the quadrotor mass (1.25 kg);
- $\mathbf{F}_{\text{thrust}} = [0, 0, \sum_{i=1}^4 T_i]^T$ is the collective rotor thrust;
- $\mathbf{D} = \text{diag}(d_x, d_y, d_z)$ is the translational aerodynamic drag tensor;
- $\mathbf{I}$ is the quadrotor moment of inertia tensor;
- $\boldsymbol{\omega} = [p, q, r]^T$ is the angular velocity in the body frame;
- $\boldsymbol{\tau}$ is the control torque vector generated by differential rotor speeds.

6.3.2 Virtual Sensor Emulation (LiDAR & Swarm Telemetry)

1. **Ray-Cast LiDAR:** Each agent projects $K = 32$ omnidirectional ray-casts (horizontal span 360° , vertical pitch $\pm 25^\circ$) up to a maximum range of $d_{\text{max}} = 30.0$ m. The normalized distance reading is:

$$s_k = \frac{\min(d_k, d_{\text{max}})}{d_{\text{max}}} \in [0.0, 1.0]$$

1. **Swarm Telemetry:** Agents within an ad-hoc communication radius $R_{\text{comm}} = 50.0$ m broadcast relative state vectors $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ and relative velocities $\Delta \mathbf{v}_{ij} = \mathbf{v}_j - \mathbf{v}_i$.
2. **Acoustic Rescue Beacon:** An omnidirectional acoustic emergency beacon is located inside a damaged structure. The acoustic intensity received by agent i follows an inverse-square attenuation law with environmental absorption coefficient μ_{air} .

$$\Phi_i(\mathbf{p}_i) = \frac{\Phi_0}{\|\mathbf{p}_i - \mathbf{p}_{\text{target}}\|^2} \cdot e^{-\mu_{\text{air}} \|\mathbf{p}_i - \mathbf{p}_{\text{target}}\|}$$

6.3.3 MAPPO Policy Architecture & Formulation

The swarm policy is trained via **Multi-Agent Proximal Policy Optimization (MAPPO)** using a shared policy network with decentralized observations:

1. Local Observation Space ($o_i^t \in \mathbb{R}^{58}$):

- Agent linear velocity: $\mathbf{v}_i \in \mathbb{R}^3$
- Agent orientation (quaternion): $\mathbf{q}_i \in \mathbb{R}^4$
- Agent angular velocity: $\boldsymbol{\omega}_i \in \mathbb{R}^3$
- 32-beam normalized LiDAR depth: $\mathbf{s}_i \in \mathbb{R}^{32}$
- Acoustic signal strength and temporal gradient: $[\Phi_i, \Delta\Phi_i] \in \mathbb{R}^2$
- K-nearest swarm neighbors relative offsets: $[\Delta\mathbf{p}_{i,1}, \dots, \Delta\mathbf{p}_{i,4}] \in \mathbb{R}^{12}$
- Minimum distance to nearest detected obstacle: $d_{\text{obs}} \in \mathbb{R}^1$
- Remaining mission time fraction: $\tau \in \mathbb{R}^1$

1. Continuous Action Space ($a_i^t \in \mathbb{R}^4$):

- Normalized collective thrust command: $u_T \in [-1.0, 1.0]$
- Body roll rate command: $\omega_{x,\text{cmd}} \in [-1.0, 1.0]$
- Body pitch rate command: $\omega_{y,\text{cmd}} \in [-1.0, 1.0]$
- Body yaw rate command: $\omega_{z,\text{cmd}} \in [-1.0, 1.0]$

1. PPO Clipped Objective:

$$\mathcal{L}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|o_t)}{\pi_{\theta_{\text{old}}}(a_t|o_t)}$ and $\hat{A}_t$ is the Generalized Advantage Estimator (GAE) computed by the centralized critic $V_\phi(s_t)$.

6.3.4 Multi-Objective Reward Function Formulation

The global reward R_i^t balances target exploration, team cohesion, and collision avoidance:

$$R_i^t = R_{\text{beacon}} + R_{\text{cohesion}} + R_{\text{col}} + R_{\text{hazard}} + R_{\text{energy}} + R_{\text{step}}$$

Component	Mathematical Definition	Operational Meaning
Beacon Progress	$R_{\text{beacon}} = \lambda_1 (\Phi_i^t - \Phi_i^{t-1}) + \mathbf{1}_{\{\ \mathbf{p}_i - \mathbf{p}_{\text{target}}\ < r_{\text{found}}\}} \cdot R_{\text{success}}$	Rewards climbing acoustic intensity; massive bonus (+500) upon target localization.
Swarm Cohesion	$R_{\text{cohesion}} = -\lambda_2 \sum_{j \neq i} \max(0, \ \mathbf{p}_i - \mathbf{p}_j\ - d_{\text{max_sep}})^2$	Penalizes agents straying beyond communication radius.
Inter-Agent Repulsion	$R_{\text{repulse}} = -\lambda_3 \sum_{j \neq i} \max(0, d_{\text{safe}} - \ \mathbf{p}_i - \mathbf{p}_j\)^2$	Strong potential repulsion preventing mid-air swarm collisions.
Building Obstacle Collision	$R_{\text{col}} = -200 \cdot \mathbf{1}_{\text{Collision}(\text{Building})} - \lambda_4 \max(0, d_{\text{clearance}} - d_{\text{obs}})^2$	Severe termination penalty for structural impacts; soft boundary buffer.
Catenary Wire Hazard	$R_{\text{hazard}} = -500 \cdot \mathbf{1}_{\text{Collision}(\text{Wire})} - 25 \cdot \max(0, d_{\text{wire_crit}} - d_{\text{wire}})$	Fatal penalty for powerline contact; induces strong upward/downward clearance maneuver.
Kinematic Effort	$R_{\text{energy}} = -\lambda_5 \ \mathbf{u}_t\ ^2$	Penalizes aggressive, jittery actuator usage to conserve battery in real flight.
Step Penalty	$R_{\text{step}} = -0.1$	Constant negative penalty encouraging time-optimal mission trajectories.

7. Data, Computational Infrastructure & Software Stack

7.1 Remote Sensing & GIS Data Assets

- **Satellite Source:** SpaceNet 2 (AOI 2 — Las Vegas), 0.3 m WorldView-3 optical RGB imagery.
- **Topographic Base:** AWS Open Data Terrarium Digital Elevation Models (10 m–30 m resolution).
- **Metadata Sources:** Ephemeris XML sidecars providing solar zenith/azimuth angles (θ_s, ϕ_s) and off-nadir angles.
- **Format Architecture:** Cloud-Optimized GeoTIFFs (COG), GeoJSON feature collections (EPSG:4326 / EPSG:32611 UTM Zone 11N).

7.2 Hardware Compute Environment

- **Local Workstation:** NVIDIA GeForce GTX 1660 Super (6 GB VRAM), 6-core Intel Core i5 / AMD Ryzen, 32 GB RAM, 2 TB NVMe SSD.
- **Compute Optimization Strategy:**
 - Mask2Former / SAM 2 inference utilizing mixed precision (fp16 / bf16).
 - Headless batch processing in Blender.

- Sub-region vectorized tiling: The 4.0 km^2 study corridor is divided into $512 \text{ m} \times 512 \text{ m}$ operational cells to preserve 60 FPS in Unreal Engine 5.

7.3 Software Ecosystem & Libraries

Layer	Technology / Framework	Role in Project
Foundation Vision	Mask2Former (HuggingFace / PyTorch 2.5)	Universal
Transformer instance segmentation		
Edge Refinement	SAM 2 (Meta AI Research)	Zero-shot
promptable boundary snapping		
Geospatial Math	GDAL, Rasterio, GeoPandas, Shapely	CRS transforms,
DP regularization, nDSM		
Micro-Hazard Engine	NumPy, SciPy	SAVI algebra,
catenary curve fitting		
3D Pipeline	Blender 4.2 LTS (`bpy` API)	Automated
triangulation, manifold repair		
Physics Simulator	Unreal Engine 5.4 (Chaos Physics)	High-fidelity
6-DOF simulation & colliders		
MADRL Engine	CleanRL / PettingZoo / PyTorch	MAPPO swarm
actor-critic networks		

8. Comprehensive Evaluation Plan

8.1 Geospatial Extraction Benchmarks

Evaluated against held-out ground truth vectors on the 146 SpaceNet test tiles:

- **Instance Metrics:** Standard COCO Average Precision (AP , AP_{50} , AP_{75}).
- **Boundary Precision:** Boundary IoU (BIOU) quantifying corner fidelity and edge sharpness:

$$\text{BIOU} = \frac{|(A_d \cap A) \cap (B_d \cap B)|}{|(A_d \cap A) \cup (B_d \cap B)|}$$

- **Height Accuracy:** Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) between estimated heights and shadow trigonometric reference values:

$$\text{RMSE}_h = \sqrt{\frac{1}{N} \sum_{i=1}^N (h_i^{\text{pred}} - h_i^{\text{shadow}})^2}$$

8.2 Physics Simulation & Collision Integrity

- **Mesh Manifoldness Rate:** Percentage of generated building meshes with zero non-manifold edges.
- **Simulation Framerate:** Continuous monitoring of UE5 frame rendering time (≤ 16.6 ms for 60 FPS) during active 8-UAV swarm flight.
- **Collision Boundary Precision:** Verification that physical Chaos colliders match visual meshes within a tolerance of < 0.10 m.

8.3 MADRL Swarm Navigation & Mission Metrics

Swarm performance will be evaluated across 500 test episodes with randomized initial spawns and target locations:

- **Mission Success Rate (MSR):** Percentage of episodes where at least one UAV enters the target radius ($r_{\text{found}} = 3.0$ m) within the time limit ($T_{\text{max}} = 120$ s).
- **Collision Frequency:**
 - Inter-agent collision rate: Percentage of episodes terminated by mid-air collision.
 - Obstacle impact rate: Total building/ground collisions per mission.
 - Wire entanglement rate: Total catenary cable contacts.
- **Swarm Energy Expenditure:** Total kinetic control effort $\int_0^T \|\mathbf{u}_t\|^2 dt$.
- **Baseline Comparative Study:** Benchmark MAPPO against:
 1. Centralized 3D A* Search (idealized benchmark with full global map knowledge).
 2. Decentralized RRT* with collision cones (classical geometric motion planning).
 3. Single-Agent Independent PPO (IPPO, treating other swarm members as dynamic environmental noise).

9. Risk Assessment & Mitigation Matrix

Risk Identified	Severity	Likelihood	Mitigation Strategy
VRAM Bottlenecks during Mask2Former / SAM 2 Inference	High	Medium	Execute inference tile-by-tile using gradient-free <code>torch.inference_mode()</code> and mixed precision (<code>fp16</code>); offload intermediate tensors to system RAM.
Shadows Occluded by Adjacent Tall Structures	Medium	High	Implement quality-check filtering: only validate shadow lengths for structures where the solar ray azimuth is unblocked by neighboring buildings.
Non-Manifold Geometry Crashing UE5 Physics Engine	High	Low	Automated Blender pre-flight script runs <code>mesh.clean_up</code> and manifold testing, automatically welding vertices $< 2\text{ mm}$ apart before export.
MADRL Exploration Collapse in Complex Urban Canyons	High	Medium	Implement Curriculum Learning: spawn agents initially near the target in open space, progressively increasing obstacle density and travel distance as policy converges.
Chaos Physics Jitter with Thin Wire Colliders	Medium	Medium	Model catenary cables as compound collision capsules with expanded conservative collision radiuses ($r = 0.40\text{ m}$) rather than sub-millimeter physics lines.

10. Work Plan, Milestone Schedule & Gantt Alignment

10.1 Academic Phases & Milestones

The project is structured across four continuous academic phases:

Phase 1: Geospatial Extraction Engine (Months 1-2)

- └─ Mask2Former fine-tuning on SpaceNet 2
- └─ SAM 2 boundary refinement & DP orthogonalization
- └─ nDSM height estimation & shadow trigonometric cross-validation

Phase 2: Micro-Hazards & Geometry Compilation (Months 3-4)

- └─ SAVI tree canopy segmentation
- └─ Catenary powerline mathematical modeling
- └─ Automated Blender asset pipeline & mesh manifold verification

Phase 3: Unreal Engine 5 Environment Setup (Months 5-6)

- └─ DEM terrain import & satellite texture projection
- └─ Chaos Physics convex collider generation
- └─ 6-DOF quadrotor flight dynamics & ray-cast LiDAR emulation

Phase 4: MADRL Swarm Policy & Benchmarking (Months 7-8)

- └─ MAPPO policy training (CTDE framework)
- └─ Multi-objective reward tuning & acoustic beacon localization
- └─ Quantitative benchmarking against A* and classical planners

Final Phase: Dissertation & Defense (Months 9-10)

- └─ Comprehensive failure mode analysis
- └─ Dissertation manuscript submission
- └─ Formal thesis oral defense

10.2 Division of Responsibilities & Team Work Breakdown

To ensure rigorous execution and balanced academic contribution across the thesis lifecycle, engineering and research responsibilities are structured co-equally between the co-investigators:

Research Domain / Engineering Module	Primary Lead	Supporting Co-Investigator	Key Responsibilities & Deliverables
Stage 1: Satellite Extraction & Instance Segmentation	Haider Taqveen	Mashaf Majeed	Fine-tuning Mask2Former on SpaceNet 2; SAM 2 zero-shot edge refinement; Douglas-Peucker orthogonalization pipeline; extraction metric validation (IoU, BIoU, AP).
Stage 2: Topography, nDSM & Micro-Hazard Modeling	Mashaf Majeed	Haider Taqveen	Stereoscopic DSM ground filtering to compute nDSM; solar shadow trigonometric verification ($h = L_{\text{shadow}} \tan \theta_s$); SAVI vegetation canopy masking ($L = 0.5$); catenary powerline mathematical modeling ($y = a \cdot [\cosh(x/a) - 1]$).
Stage 3: 3D Mesh Manifolding & Physics Simulator	Haider Taqveen	Mashaf Majeed	Blender Python (<code>bpy</code>) automated asset generation; watertight manifold repair (≤ 2 mm vertex welding); Unreal Engine 5 Chaos Physics convex hull and capsule collider integration; framerate optimization (≥ 60 FPS).
Stage 4: Multi-Agent DRL Swarm Engine	Mashaf Majeed	Haider Taqveen	MAPPO multi-agent reinforcement learning architecture (CTDE); 6-DOF quadrotor kinematic constraints; 32-beam LiDAR emulation; multi-objective reward formulation (acoustic rescue beacon, collision & wire entanglement penalties).
Comparative Benchmarking & Thesis Manuscript	Haider Taqveen & Mashaf Majeed	Dr. Munawar Ali Shah (Supervisor)	Comparative benchmarking against centralized 3D A and decentralized RRT; trajectory safety analysis; comprehensive thesis dissertation writeup and oral defense preparation.

11. Expected Academic Deliverables & Contributions

11.1 Tangible Deliverables

- Fully Automated Geospatial Reconstruction Pipeline (`src/data_prep/` , `src/geometry/`):** Python toolchain converting raw optical satellite tiles into cadastre-grade 3D meshes with shadow-validated heights.
- Compiled Unreal Engine 5 Digital Twin:** A high-fidelity, georeferenced simulation environment of Las Vegas Sector 7 featuring Chaos Physics colliders, micro-hazards, and terrain topography.
- MADRL Swarm Policy Repository (`src/models/marl_swarm/`):** Clean, modular PyTorch implementation of MAPPO governing 6-DOF quadrotors equipped with virtual LiDAR and acoustic localization heads.

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4. **Interactive Exploratory Notebooks (notebooks/):** Reproducible validation notebooks demonstrating extraction metrics, height error bounds, and swarm trajectory visualizations.
 5. **Final BS Thesis Manuscript & Oral Defense Presentation.**

11.2 Key Scientific Contributions

- **Bridging GIS and Real-Time Autonomous Swarm Simulation:** Demonstrates that foundation models (Mask2Former + SAM 2) combined with classical photogrammetric shadow trigonometry can generate simulation-ready physics environments without manual 3D modeling.
 - **Empirical Demonstration of Micro-Hazard Impact on Swarms:** Proves the necessity of explicit physical modeling for catenary wires and vegetation in urban aerial swarm simulation.
 - **Validated MADRL Swarm Policy:** Establishes benchmark convergence rates and trajectory safety metrics for decentralized swarms operating under 6-DOF constraints in photogrammetrically grounded urban digital twins.
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12. Ethical, Dual-Use & Environmental Considerations

- **Civic & Humanitarian Mission Alignment:** The primary operational focus of this research is post-disaster search and rescue, emergency logistics, and civilian infrastructure inspection. The thesis strictly rejects developing targeting algorithms, kinetic tracking, or offensive weaponization systems.
 - **Privacy Compliance:** All imagery used originates from publicly accessible SpaceNet commercial datasets released under permissive research licenses. Spatial resolution (0.3m GSD) is sufficient to identify structures and vehicles, but is incapable of resolving human faces or private personal information.
 - **Carbon Footprint & Energy Conservation:** By conducting training inside a bounded geospatial region (4.0 km²) and utilizing lightweight, sample-efficient algorithms (MAPPO), computational requirements are minimized, enabling full reproduction on local consumer hardware without wasteful data-center energy expenditure.
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13. References

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14. Appendices

Appendix A: Derivation of Shadow Trigonometry Equation

Consider an optical satellite sensor observing a vertical building wall of height h . Let θ_s denote the solar elevation angle (altitude above the local horizon), such that the angle between the solar ray vector and the vertical surface normal is $\frac{\pi}{2} - \theta_s$.

Assuming the ground datum is locally horizontal:

$$\tan(\theta_s) = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{h}{L_{\text{shadow}}}$$

Rearranging directly yields:

$$h = L_{\text{shadow}} \cdot \tan(\theta_s)$$

When accounting for sensor viewing geometry (sensor zenith θ_v and relative azimuth $\Delta\phi = \phi_s - \phi_v$), the apparent shadow length L_{app} in the image plane is corrected via projective perspective geometry:

$$L_{\text{true}} = \frac{L_{\text{app}}}{\cos(\theta_v) (1 + \tan(\theta_v) \cot(\theta_s) \cos(\Delta\phi))}$$

Appendix B: Catenary Sag Mathematical Formulation

A flexible wire suspended under uniform gravity satisfies the differential equation:

$$\frac{d^2 y}{dx^2} = \frac{1}{a} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}, \quad a = \frac{T_0}{\mu g}$$

Letting $u = \frac{dy}{dx}$, separation of variables yields:

$$\int \frac{du}{\sqrt{1+u^2}} = \int \frac{dx}{a} \implies \text{arsinh}(u) = \frac{x}{a} + C_1$$

Setting the origin at the lowest sag point where $\left.\frac{dy}{dx}\right|_{x=0} = 0$, we have $C_1 = 0$. Integrating once more:

$$y(x) = a \int \sinh\left(\frac{x}{a}\right) dx = a \cosh\left(\frac{x}{a}\right) + C_2$$

Setting $y(0) = 0 \implies C_2 = -a$, giving the standard catenary curve equation:

$$y(x) = a \left[\cosh\left(\frac{x}{a}\right) - 1 \right]$$

Appendix C: Glossary of Technical Abbreviations

Abbreviation	Expanded Technical Terminology
AOI	Area of Interest
CTDE	Centralized Training with Decentralized Execution
Dec-POMDP	Decentralized Partially Observable Markov Decision Process
DP	Douglas-Peucker (Polygon Regularization Algorithm)
DSM	Digital Surface Model
DTM	Digital Terrain Model
GSD	Ground Sampling Distance
LiDAR	Light Detection and Ranging
LWC	Large World Coordinates (64-bit Floating-Point World Origin)
MADRL	Multi-Agent Deep Reinforcement Learning
MAPPO	Multi-Agent Proximal Policy Optimization
nDSM	Normalized Digital Surface Model ($DSM - DTM$)
NDVI	Normalized Difference Vegetation Index
PPO	Proximal Policy Optimization
SAM 2	Segment Anything Model 2 (Meta AI Foundation Model)
SAVI	Soil-Adjusted Vegetation Index
UE5	Unreal Engine 5
URP	Universal Render Pipeline (Legacy Reference)